Optimal Energy Management of a Hydrogen–Electric Aircraft Propulsion System

Stiff ODE Modelling, Collocation, and Solver Comparison

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Abstract

This study uses numerical optimization methods from MECH 568 and stiff ODE modeling to examine optimal energy management for a simpler hydrogen-electric aircraft powertrain. A three-state stiff initial value problem driven by a specified mission power profile is used to write a lumped-parameter model of a PEM fuel cell stack and Thevenin battery. To measure hydrogen consumption, battery state of charge (SOC), and peak current, MATLAB's stiff solver `ode15s` (BDF family) simulates a baseline controller that allocates a set fraction of propulsive power to the fuel cell. The energy management problem is then posed as a continuous-time optimal control problem with a running cost that trades hydrogen consumption against battery current, and a terminal SOC constraint. A direct collocation discretization with trapezoidal (Crank–Nicolson) defects and `fmincon` is used to compute an approximate optimal power split. For the representative mission, the collocation strategy increases hydrogen usage from $\approx 0.75\,\mathrm{kg}$ to $\approx 1.06\,\mathrm{kg}$, but improves minimum SOC from 0.135 to 0.555 and reduces peak current from about $84\,\mathrm{A}$ to $66\,\mathrm{A}$. The outcomes show how stiff solvers and collocation-based optimal control may be used for hybrid hydrogen-electric propulsion models and clearly show a trade-off between fuel efficiency and battery stress.

Keywords: Simulation, Modelling, Optimal Control, Stiff ODEs, Hydrogen–Electric Propulsion, Numerical Methods

1 Introduction

A viable route to low-emission regional aircraft is hydrogen-electric propulsion [1]. Typically, a lithium-ion battery provides peak shaving and transient responsiveness, while a polymer electrolyte membrane (PEM) fuel cell stack provides the majority of the average propulsive power. When compared to kerosene turbines, the combined system can significantly lower CO_2 and NO_x emissions, but only provided hydrogen consumption is carefully controlled and battery current stress is kept within safe bounds throughout numerous charge-discharge cycles.

- This project builds directly on topics from MECH 568, including stiff ODE time-stepping, collocation, and nonlinear programming methods.

Such a hybrid system's functioning is essentially an energy management issue, closely connected to methods used by electric and hybrid vehicles [2]. The controller must determine how to divide power between the fuel cell and the battery for a given mission power profile $P_{\mathrm{prop}}(t)$. While favoring the battery saves hydrogen

temporarily but may result in high currents, significant depth-of-discharge swings, and thermal loading, favoring the fuel cell decreases battery degradation but increases hydrogen consumption. Even for a single typical mission, quantifying this tradeoff demands a dynamic model that takes into account the stiff electrical dynamics of the hybrid powertrain, device limitations, and power balancing.

The hybrid model presents a number of numerical challenges that are in line with MECH 568: Computational Methods for Mechanical Engineering. The power split can be expressed as a continuous-time control input, and the governing equations can be expressed as a stiff system of ordinary differential equations (ODEs). Robust ODE integrators and a limited nonlinear optimizer are needed to solve the resulting optimal control problem. This is an informative testbed for stiff solvers and collocation-based optimum control methods [3, 4], since the stiffness of the dynamics and the coupling between states and controls render naive explicit approaches inefficient or unstable.

This project addresses the following questions for a simplified hydrogen–electric aircraft powertrain and a prescribed mission profile:

RQ1 How do various power split algorithms impact peak battery current and overall hydrogen consumption for a fixed mission power profile?

RQ2 In this stiff ODE system, how does an optimal control formulation based on collocation and `fmincon` compare to a more straightforward shooting-based method?

RQ3 How sensitive are the outcomes to the cost functional's relative weighting of battery current against hydrogen usage?

RQ4 Regarding convergence behavior, stability, and computational cost, what realistic numerical trade-offs exist between the proposed solvers?

The contributions of this report are:

- I generate and implement a lumped-parameter model of a lithium-ion battery and PEM fuel cell stack connected by a predetermined propulsive power demand profile, and I describe the dynamics as a stiff ODE system.
- In order to balance hydrogen consumption against battery current stress under power, state-of-charge, and current limits, I formulate a power split optimum control problem.
- I use `fmincon` to solve two numerical solution strategies: a lower-dimensional shooting-based technique and a direct collocation strategy, and I compare how well they perform on the same mission.
- In addition to discussing useful suggestions for modeling and developing comparable hybrid hydrogen-electric powertrains, I offer numerical results that quantify the trade-off between hydrogen consumption and peak battery current for the selected mission.

2 Methods

2.1 Constants and Reporting Level

I utilize a single representative mission profile and a set of predefined parameters for the battery and fuel cell stack to keep the conversation on topic. Appendix B contains detailed numerical values; here, I provide an overview of the reporting level and conventions utilized throughout the study.

- **Mission profile.** Over a mission of duration T, the propulsive power requirement $P_{\text{prop}}(t)$ is dictated (e.g., climb–cruise–descent phases). Power is expressed in watts, and time is expressed in seconds.
- **Fuel cell stack.** A fixed number of cells N_{cell}, active area A_{fc}, and polarization parameters that specify the voltage–current relation $V_{\text{fc}}(i_{\text{fc}})$ define the PEM fuel cell stack. The lower heating value (LHV) of hydrogen or an equivalent efficiency-based approximation [5] is used to calculate hydrogen consumption, which is expressed in grams and derived from Faraday's law.
- **Battery.** A nominal capacity C_{bat} (A,"s), an open-circuit voltage function $V_{\text{OC}}(\text{SOC})$, and internal resistance parameters are used to simulate the lithium-ion battery. Battery current is expressed in amperes, while battery state of charge (SOC) is expressed as a fraction in $[0, 1]$.
- **Efficiencies.** Constant efficiencies η_{fc} and η_{bat} multiply the electrical powers from the fuel cell and battery, respectively, to represent converter and inverter losses.
- **Numerical tolerances.** ODE integrations use `ode15s` with the absolute tolerance AbsTol and relative tolerance RelTol specified in Appendix B. Using its interior-point approach and default constraint tolerances, `fmincon` solves all optimum control problems, with any deviations clearly recognized.
- **Reporting of performance metrics.**The total mass used throughout the flight is used to report hydrogen utilization. The greatest battery current and the lowest SOC reached are the main indicators of battery stress. I provide relative changes to a baseline power split approach when comparing strategies.

2.2 Physical and Mathematical Model

A PEM fuel cell stack, a lithium-ion battery, and an electric propulsion load that requires a certain power $P_{\text{prop}}(t)$ throughout the mission make up the hybrid powertrain's three primary parts. A power split instruction $u(t) \in [0, 1]$ that indicates the portion of the required propulsive power to be provided by the fuel cell serves as the control input. The battery supplies the remaining power, subject to current and state-of-charge limitations.

State variables and control input

I collect the dynamic states into a vector

$$x(t) = \begin{bmatrix} \text{SOC}(t) \\ v_{\text{rc}}(t) \\ P_{\text{fc}}(t) \end{bmatrix}, \tag{1}$$

where $\text{SOC}(t)$ is the battery state of charge, $v_{\text{rc}}(t)$ is the voltage across a simple first-order RC element used to capture battery transient behavior, and $P_{\text{fc}}(t)$ is the actual electrical power delivered by the fuel cell stack. The control input is

$$u(t) \in [0, 1], \tag{2}$$

which is interpreted as the commanded fraction of propulsive power to be supplied by the fuel cell.

Given the mission profile $P_{\text{prop}}(t)$, the commanded fuel cell power is

$$P_{\text{fc,cmd}}(t) = u(t)\, P_{\text{prop}}(t), \tag{3}$$

and the commanded battery power is

$$P_{\mathrm{bat}}(t) = \frac{P_{\mathrm{prop}}(t) - P_{\mathrm{fc}}(t)}{\eta_{\mathrm{bat}}}, \tag{4}$$

where η_{bat} is the effective battery-side conversion efficiency. Equation (4) enforces power balance between the fuel cell, the battery, and the propulsive load.

Fuel cell model and hydrogen consumption

The fuel cell stack is modeled using a static polarization relation between stack voltage and current. For a given stack current i_{fc}, the stack voltage is approximated as

$$V_{\mathrm{fc}}(i_{\mathrm{fc}}) = E_0 - A \ln(i_{\mathrm{fc}} + i_{\mathrm{d}}) - R_{\mathrm{ohm}}\, i_{\mathrm{fc}}, \tag{5}$$

where E_0 is the open-circuit voltage, A is an empirical activation loss constant, i_{d} is a small regularization current to avoid the logarithmic singularity, and R_{ohm} is the effective ohmic resistance. This form is representative of standard semi-empirical PEM stack models [5]. The electrical power delivered by the stack is

$$P_{\mathrm{fc}}(t) = \eta_{\mathrm{fc}}\, V_{\mathrm{fc}}(i_{\mathrm{fc}})\, i_{\mathrm{fc}}, \tag{6}$$

with η_{fc} representing conversion efficiency.

To capture the finite response time of the stack and its balance-of-plant, I treat the fuel cell power as a first-order lag of the commanded power:

$$\tau_{\mathrm{fc}}\, \dot{P}_{\mathrm{fc}}(t) = -P_{\mathrm{fc}}(t) + P_{\mathrm{fc,cmd}}(t), \tag{7}$$

where τ_{fc} is a time constant. Given $P_{\mathrm{fc}}(t)$ and the instantaneous voltage, the stack current is

$$i_{\mathrm{fc}}(t) = \frac{P_{\mathrm{fc}}(t)}{\eta_{\mathrm{fc}}\, V_{\mathrm{fc}}(i_{\mathrm{fc}})}. \tag{8}$$

In practice, equation (8) is solved numerically inside the right-hand side evaluation.

The hydrogen mass flow rate is approximated directly from the fuel cell power using a fixed overall stack efficiency η_{stack} and the lower heating value of hydrogen $\mathrm{LHV}_{\mathrm{H_2}}$:

$$\dot{m}_{\mathrm{H_2}}(t) = \frac{P_{\mathrm{fc}}(t)}{\eta_{\mathrm{stack}}\, \mathrm{LHV}_{\mathrm{H_2}}}, \tag{9}$$

and the total hydrogen consumed over the mission is

$$m_{\mathrm{H_2,tot}} = \int_0^T \dot{m}_{\mathrm{H_2}}(t)\, \mathrm{d}t. \tag{10}$$

This matches the implementation used in the MATLAB code, where the power-based estimate is convenient and consistent with the parameter tables in Appendix B.

Battery model

The battery is represented by a simple Thevenin model: an open-circuit voltage source $V_{\mathrm{OC}}(\mathrm{SOC})$ in series with a resistance R_0 and a first-order RC branch (R_1, C_1) that captures transient polarization. The terminal voltage is

$$V_{\mathrm{bat}}(t) = V_{\mathrm{OC}}(\mathrm{SOC}(t)) - R_0\, i_{\mathrm{bat}}(t) - v_{\mathrm{rc}}(t), \tag{11}$$

and the RC voltage evolves according to

$$\tau_{\mathrm{rc}}\, \dot{v}_{\mathrm{rc}}(t) = -v_{\mathrm{rc}}(t) + R_1\, i_{\mathrm{bat}}(t), \tag{12}$$

where $\tau_{\mathrm{rc}} = R_1 C_1$.

Given the battery power $P_{\mathrm{bat}}(t)$ from equation (4), the battery current $i_{\mathrm{bat}}(t)$ satisfies

$$P_{\mathrm{bat}}(t) = V_{\mathrm{bat}}(t)\, i_{\mathrm{bat}}(t). \tag{13}$$

Substituting equation (11) into equation (13) yields a quadratic equation for $i_{\mathrm{bat}}(t)$,

$$R_0\, i_{\mathrm{bat}}^2(t) - \big(V_{\mathrm{OC}}(\mathrm{SOC}(t)) - v_{\mathrm{rc}}(t)\big)\, i_{\mathrm{bat}}(t) + P_{\mathrm{bat}}(t) = 0, \tag{14}$$

which is solved at each time step, and the physically relevant root is selected based on the sign of $P_{\mathrm{bat}}(t)$. This Thevenin structure is widely used for lithium-ion batteries and can reproduce both runtime and I–V characteristics with modest complexity [6].

The state-of-charge dynamics follow from charge balance:

$$\dot{\mathrm{SOC}}(t) = -\frac{i_{\mathrm{bat}}(t)}{C_{\mathrm{bat}}}, \tag{15}$$

with C_{bat} the effective capacity. The SOC is constrained to lie within prescribed bounds

$$\mathrm{SOC}_{\min} \le \mathrm{SOC}(t) \le \mathrm{SOC}_{\max}, \tag{16}$$

and the battery current magnitude is limited by

$$|i_{\mathrm{bat}}(t)| \le i_{\mathrm{bat,max}}. \tag{17}$$

ODE system form

Collecting equations (7), (12), and (15) yields the stiff ODE system

$$\dot{x}(t) = f\big(x(t), u(t), t\big), \tag{18}$$

Hence, given the current power split command $u(t)$ and mission power profile $P_{\mathrm{prop}}(t)$, the right-hand side assesses the fuel cell power dynamics, the RC voltage dynamics, and the SOC dynamics. The fuel cell and battery currents are recovered by numerically solving the algebraic relations (8), (10), and (14) within $f(\cdot)$. The dynamical core of the optimum control issue examined in the rest of the paper is formed by this ODE system and the inequality constraints (16)–(17).

2.3 Baseline Stiff ODE Simulation

For any prescribed power split strategy $u(t)$ the hybrid powertrain is governed by the stiff ODE system

$$\dot{x}(t) = f\big(x(t), u(t), t\big), \tag{19}$$

where $x(t) = [\mathrm{SOC}(t),\, v_{\mathrm{rc}}(t),\, P_{\mathrm{fc}}(t)]^\top$ and $f(\cdot)$ is defined by equations (7), (12), (15), and (18). The mission profile $P_{\mathrm{prop}}(t)$ is treated as a known, piecewise-constant input constructed from a small number of power levels and segment durations (Appendix B).

MATLAB's stiff solver `ode15s` is used for all forward simulations in this work, including the baseline and the postprocessing of optimal solutions. Variable-order backward differentiation formula (BDF) schemes, which are implicit, A-stable multistep techniques closely linked to the Backward Euler and trapezoid (Crank–Nicolson) time-stepping schemes covered in the MECH 568 on stability and stiff IVPs, provide the basis of the solver `ode15s`. Unless otherwise noted, the solver is set up with a maximum step size of $5\,\mathrm{s}$, absolute tolerance $\texttt{AbsTol} = 10^{-8}$, and relative tolerance $\texttt{RelTol} = 10^{-6}$. These parameters are in line with conventional recommendations for stiff problems [3] and balance computational cost over the $1800\,\mathrm{s}$ mission horizon with robustness on the stiff RC and fuel cell dynamics.

For the baseline case I consider a constant power split $u(t) \equiv u_0 = 0.6$, meaning that the fuel cell is commanded to supply 60% of the instantaneous propulsive power demand and the battery supplies the remaining fraction. The initial state is chosen as

$$x(0) = \begin{bmatrix} \mathrm{SOC}_0 \\ 0 \\ u_0\, P_{\mathrm{prop}}(0) \end{bmatrix}, \tag{20}$$

with $\mathrm{SOC}_0 = 0.8$ representing a nearly full battery, zero initial RC voltage, and fuel cell power consistent with the initial split. After integration, the time histories are postprocessed to compute the battery current, hydrogen mass flow, and derived performance metrics such as total hydrogen consumption $m_{\mathrm{H_2,tot}}$, minimum SOC, and peak current. As a basic consistency check, simulations are discarded if they violate hard constraints such as negative hydrogen mass or solver failures (e.g. repeated step rejections).

2.4 Optimal Control Formulation

A continuous-time optimal control problem is used to represent the energy management issue. With state $x(t) \in \mathbb{R}^3$ and scalar control $u(t) \in [0, 1]$, the stiff ODE system $\dot{x}(t) = f(x(t), u(t), t)$ presented in Section 2.2 provides the state dynamics. The control shows the percentage of propelling power that the fuel cell is instructed to provide. The set of acceptable controls is

$$\mathcal{U} = \{u : [0, T] \to [0, 1]\}, \tag{21}$$

so that the fuel cell is never asked to deliver negative power or more than the entire propulsive demand.

The objective functional combines hydrogen usage with a quadratic penalty on battery current,

$$J[u(\cdot)] = \int_0^T \Big(w_H\, \dot{m}_{\mathrm{H_2}}(t) + w_I\, i_{\mathrm{bat}}^2(t)\Big)\, \mathrm{d}t, \tag{22}$$

where $i_{\mathrm{bat}}(t)$ is the battery current derived from the Thevenin model in Section 2.2 and $\dot{m}_{\mathrm{H_2}}(t)$ is the instantaneous hydrogen mass flow rate defined in equation (10). The relative priority of reducing hydrogen

consumption against limiting battery current stress is controlled by the weights $w_H > 0$ and $w_I > 0$. In this work, I concentrate on the representative option $(w_H, w_I) = (1, 10^{-4})$, which is comparable to energy-management formulas for hybrid vehicles [2].

The optimization is subject to dynamic, path, and endpoint constraints. The dynamic constraints are simply the ODEs,

$$\dot{x}(t) = f(x(t), u(t), t), \qquad x(0) = x_0, \tag{23}$$

where x_0 is given by the baseline initial condition. Path constraints capture SOC and current limits,

$$\mathrm{SOC}_{\min} \le \mathrm{SOC}(t) \le \mathrm{SOC}_{\max}, \quad |i_{\mathrm{bat}}(t)| \le i_{\mathrm{bat,max}}, \qquad 0 \le t \le T, \tag{24}$$

and fuel cell power bounds $P_{\mathrm{fc,min}} \le P_{\mathrm{fc}}(t) \le P_{\mathrm{fc,max}}$ are enforced inside the dynamics. Finally, a terminal constraint ensures that the battery does not end the mission nearly depleted:

$$\mathrm{SOC}(T) \ge \mathrm{SOC}_{\mathrm{target}}, \tag{25}$$

with $\mathrm{SOC}_{\mathrm{target}} = 0.6$ in the nominal case. Intuitively, larger values of w_I and more restrictive SOC/current constraints lead to battery-friendly strategies that rely more heavily on hydrogen, whereas larger w_H or looser constraints favour hydrogen-efficient but battery-stressing strategies.

2.5 Collocation Implementation

To solve the optimal control problem numerically I adopt a direct collocation approach. The time interval $[0, T]$ is partitioned into $N = 40$ uniform segments with nodes $t_k = kT/N$, $k = 0, \ldots, N$. The state and control are approximated at these nodes by decision variables

$$\{x_k\}_{k=0}^{N}, \qquad \{u_k\}_{k=0}^{N}, \tag{26}$$

so that the full decision vector has dimension $4(N + 1) = 164$ in the present implementation. Between nodes, the state is assumed to vary linearly and the dynamics are enforced using a trapezoidal collocation scheme [4].

For each segment $[t_k, t_{k+1}]$ with step size Δt_k, the collocation defect is

$$x_{k+1} - x_k - \frac{\Delta t_k}{2}\big(f(x_k, u_k, t_k) + f(x_{k+1}, u_{k+1}, t_{k+1})\big) = 0, \tag{27}$$

which is imposed as a nonlinear equality constraint for $k = 0, \ldots, N - 1$. The trapezoidal collocation condition $x_{k+1} - x_k - \frac{\Delta t_k}{2}\big(f_k + f_{k+1}\big) = 0$ is mathematically equivalent to applying the Crank–Nicolson / trapezoid time-stepping method to the ODE system, so the stability properties mirror those of the second-order A-stable schemes covered in the time-stepping lecture.

The initial-node equality $x_0 = x(0)$ enforces the initial condition, while the terminal SOC constraint and battery current limits are imposed as inequalities at the nodes. The running cost is discretized using the same trapezoidal rule as in Section 3.2,

$$J \approx \sum_{k=0}^{N-1} \frac{\Delta t_k}{2}\Big(w_H\big(\dot{m}_{\mathrm{H}_2,k} + \dot{m}_{\mathrm{H}_2,k+1}\big) + w_I\big(i_{\mathrm{bat},k}^2 + i_{\mathrm{bat},k+1}^2\big)\Big), \tag{28}$$

with $\dot{m}_{\mathrm{H}_2,k}$ and $i_{\mathrm{bat},k}$ obtained from the algebraic relations at each node using the same functions as in the forward simulation.

With the interior-point algorithm, variable bounds $0 \leq u_k \leq 1$ and $\mathrm{SOC}_{\min} \leq \mathrm{SOC}_k \leq \mathrm{SOC}_{\max}$, and the same numerical tolerances used in Section 3.4, the resultant nonlinear program is fed into MATLAB's `fmincon`. Using $u_k \equiv 0.6$ and straightforward linear interpolants for SOC_k, $v_{\mathrm{rc},k}$, and $P_{\mathrm{fc},k}$, the initial estimate is built from the constant-split baseline trajectory. Following optimization, a high-fidelity trajectory for postprocessing hydrogen usage and current metrics is obtained by interpolating the optimal power split $\{u_k^\star\}$ in time and passing it back through `ode15s`.

2.6 Shooting-Based Approach

Treating the stiff ODE solver as a black box and defining the power split with a limited number of design variables is a conceptual substitute for direct collocation. The control $u(t)$ is expressed as a piecewise-constant function with a few segments in a straightforward shooting-based formulation, such as

$$u(t) = \begin{cases} u_1, & t_0 \leq t < t_1, \\ u_2, & t_1 \leq t < t_2, \\ \vdots & \\ u_M, & t_{M-1} \leq t \leq T, \end{cases} \tag{29}$$

where $u_1, \ldots, u_M$ are decision variables bounded between 0 and 1 and the breakpoints t_i are chosen a priori (for example aligned with mission segments). For any candidate vector $u = [u_1, \ldots, u_M]^\top$ the stiff ODE system is integrated forward using `ode15s`, and the objective J and constraint violations are evaluated from the resulting trajectory.

This results in a low-dimensional nonlinear program in the M parameters, which may also be solved using a derivative-free optimizer or `fmincon`. Because the forward simulation code is directly reused, a shooting strategy is simpler to build and has significantly lower individual function evaluations than collocation [4]. However, it can be more sensitive to the choice of parameterization and starting assumption, particularly for stiff dynamics, and it offers less direct control over state inequality limitations like SOC and current limits. I utilize the shooting formulation mostly as a conceptual point of comparison in the discussion and concentrate on the collocation solution as the primary numerical result for this project.

2.7 Assumptions

The modelling and numerical results in this report rely on several simplifying assumptions that are worth stating explicitly:

- **Prescribed mission profile.** Regardless of the power split approach selected, it is believed that the propulsive power demand $P_{\mathrm{prop}}(t)$ is known and fixed beforehand. Only the electrical power profile is taken into account; flight mechanics, speed fluctuations, and altitude impacts are disregarded.
- **Lumped-parameter component models.** Lumped-parameter models with constant polarization and resistance parameters are used to represent both the battery and the fuel cell stack. Aging effects, water management, detailed gas transport, and temperature dynamics are disregarded [5, 6].
- **Constant efficiencies.** Constant efficiencies η_{fc} and η_{bat} that are independent of load or operating point capture converter and inverter losses. The hydrogen mass calculation uses the total stack efficiency η_{stack} to account for the fuel cell's balance-of-plant power consumption.

- **Battery model simplifications.** A single first-order RC network represents the internal dynamics, and the battery open-circuit voltage is approximated as an affine function of SOC. Temperature dependency, hysteresis, and rate-dependent capacity loss are disregarded [6].
- **Numerical settings.** Every simulation is carried out in MATLAB with double precision, utilizing the `ode15s` and `fmincon` options as detailed in Sections 2.3 and 2.5. No adaptive mesh refinement is employed; the collocation grid is predetermined [4].
- **Single-trajectory focus.** One baseline algorithm and one collocation-derived solution for a single mission profile and weight pair (w_H, w_I) form the basis of the tradeoff analysis. Variations in the mission and parameter uncertainty are not investigated.

In the context of this course, these simplifications are suitable for an exploratory computational research; but, before utilizing the model for extensive design or certification studies, they should be reviewed.

3 Results

3.1 Baseline Mission Simulation

Before solving any optimal control problem, I first examine the behaviour of the hybrid powertrain under a simple heuristic power split. For the baseline case the fuel cell is commanded to carry a fixed fraction $u_0 = 0.6$ of the propulsive power demand, so that $P_{\mathrm{fc,cmd}}(t) = u_0 P_{\mathrm{prop}}(t)$ and the battery supplies the remaining power through equation (4). The stiff ODE system in equation (18) is integrated with `ode15s` over the full $T = 1800\,\mathrm{s}$ mission.

The resulting time histories of propulsive power $P_{\mathrm{prop}}(t)$, fuel cell power $P_{\mathrm{fc}}(t)$, and battery power $P_{\mathrm{bat}}(t)$, along with the battery state of charge and current, are displayed in Figure 1. The power trajectories match the recommended four-segment mission profile, which consists of a high-power ascent stage, a lengthy cruise at $50\,\mathrm{kW}$, descent, and approach segments at reduced power. Due to first-order stack dynamics, the fuel cell tracks around 60% of the required power with a slight lag, and the battery carries the remaining portion.

At the completion of the task, the state-of-charge trajectory drops nearly monotonically from the starting $\mathrm{SOC}(0) = 0.8$ to $\mathrm{SOC}(T) \approx 0.135$. This indicates that the battery is severely depleted by the simple constant power split, which also goes against the intended lower bound $\mathrm{SOC}_{\min} = 0.2$ presented in Section 2.3. The battery current, on the other hand, is still quite low, with a peak magnitude of roughly $84.4\,\mathrm{A}$, much below the set limit $i_{\mathrm{bat,max}} = 200\,\mathrm{A}$. The major performance indicators used later for comparison with optimal techniques are summarized in Table 1.

Overall, the baseline simulation illustrates the trade-off that drives the remainder of this research: a naive approach that draws the battery's state of charge (SOC) too low for safe repeated operation while using a comparatively modest quantity of hydrogen for peak shaving. Redistributing the power split in time to meet the SOC requirement while managing hydrogen consumption and battery current stress is the goal of the optimum control formulations in Sections 2.4–2.6.

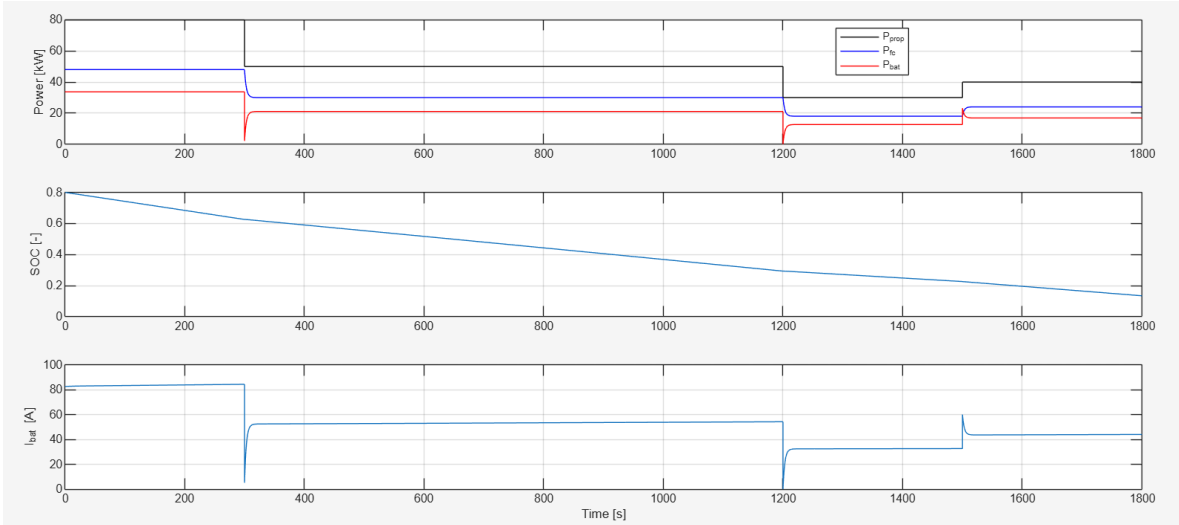

Figure 1: Baseline mission simulation with constant power split $u_0 = 0.6$. Top: propulsive power demand $P_{\text{prop}}(t)$ and the resulting fuel cell and battery powers. Middle: battery state-of-charge. Bottom: battery current time history.

Table 1: Baseline performance metrics for constant power split $u_0 = 0.6$.

Quantity	Symbol / definition	Baseline value
Total hydrogen consumed	$m_{\text{H}_2,\text{tot}} = \int_0^T \dot{m}_{\text{H}_2}(t)\,\text{d}t$	$0.751\,\text{kg}$
Peak battery current magnitude	$\max_t \lvert i_{\text{bat}}(t)\rvert$	$84.4\,\text{A}$
Minimum state of charge	$\min_t \text{SOC}(t)$	0.135
SOC lower bound (constraint)	$\text{SOC}_{\min}$	0.20
Mission duration	T	$1800\,\text{s}$

3.2 Optimal Power Split Trajectories

The direct collocation formulation described in Section 2.5 was implemented with $N = 40$ segments over the $T = 1800\,\text{s}$ mission. The decision vector collects the state values $\{\text{SOC}_k, v_{\text{rc},k}, P_{\text{fc},k}\}_{k=0}^N$ and power split commands $\{u_k\}_{k=0}^N$ at the collocation nodes. The objective functional is discretized using a trapezoidal rule,

$$J \approx \sum_{k=0}^{N-1} \frac{\Delta t_k}{2}\Big[w_H\big(\dot{m}_{\text{H}_2}(t_k) + \dot{m}_{\text{H}_2}(t_{k+1})\big) + w_I\big(i_{\text{bat}}^2(t_k) + i_{\text{bat}}^2(t_{k+1})\big)\Big], \tag{30}$$

with weights $w_H = 1$ and $w_I = 10^{-4}$. The terminal constraint $\text{SOC}(T) \geq \text{SOC}_{\text{target}}$ with $\text{SOC}_{\text{target}} = 0.6$ and the pointwise current limits $\lvert i_{\text{bat}}(t)\rvert \leq i_{\text{bat,max}}$ are enforced at the nodes. The resulting nonlinear program is solved using `fmincon` (interior-point algorithm) starting from the constant-split baseline $u(t) \equiv 0.6$.

The baseline and collocation-derived trajectories are contrasted in Figure 2. The optimal fuel cell power $P_{\text{fc}}(t)$ (blue dashed) in the power subplot is constantly greater than the baseline level that tracks 60% of the demand, especially in the latter approach and high-power ascent segments. When the fuel cell exceeds the instantaneous demand, the associated battery power $P_{\text{bat}}(t)$ (red dashed) decreases and even turns slightly

negative near some transitions, suggesting brief charging. Since the mission power profile $P_{\text{prop}}(t)$ is the same in both scenarios, it is only plotted for the baseline.

The middle subplot demonstrates how the best course of action keeps the charge in a far healthier state throughout the operation. The collocation-based policy completes the mission at $\text{SOC}(T) \approx 0.555$ and never falls below the intended operating window, whereas the baseline SOC decreases from $\text{SOC}(0) = 0.8$ to $\text{SOC}(T) \approx 0.135$. The terminal SOC restriction and the slight battery current penalty in the operating cost are directly responsible for such conduct.

The corresponding battery current trajectories are shown in the bottom subplot. The maximal magnitude of $\max_t |i_{\text{bat}}(t)| \approx 84.4\,\text{A}$ results from the baseline controller using about $80\,\text{A}$ during the climb segment and $50\,\text{A}$ to $60\,\text{A}$ during cruise. With usual values close to $40-50\,\text{A}$ and a lowered peak of roughly $66.1\,\text{A}$, the ideal technique lowers currents throughout all segments. Quantitatively, switching from the baseline to the collocation solution improves the minimum SOC from 0.135 to 0.555 and lowers the peak current by around 20% while increasing total hydrogen usage from $m_{\text{H}_2,\text{tot}} = 0.751\,\text{kg}$ to $1.062\,\text{kg}$ (about a 40% increase). This demonstrates the main trade-off of the issue: using more energy from the hydrogen tank is necessary to protect the battery from deep discharge and high current stress.

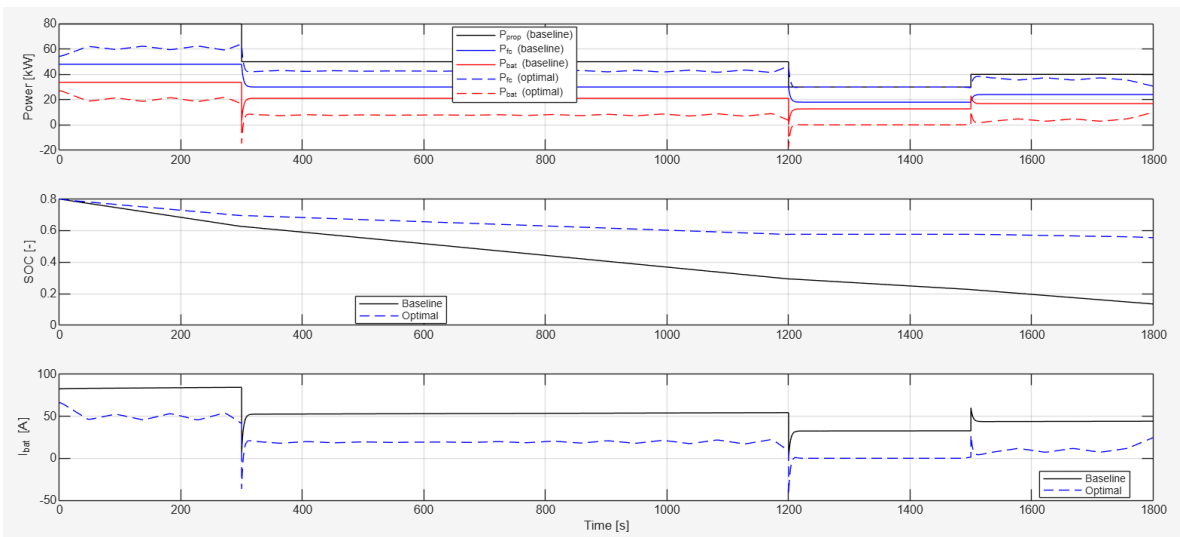

Figure 2: Baseline (solid) versus collocation-based optimal (dashed) power split. Top: propulsive power demand and the corresponding fuel cell and battery powers. Middle: battery state-of-charge trajectories. Bottom: battery current. The optimal strategy leans more heavily on the fuel cell to maintain a higher SOC and reduce current magnitude.

3.3 Tradeoff and Sensitivity Studies

A clear trade-off between hydrogen consumption and battery stress is already demonstrated by the baseline controller and the collocation-based optimal solution. While the collocation solution explicitly optimizes a running cost with weights $(w_H, w_I) = (1, 10^{-4})$ subject to a terminal SOC limit, the baseline heuristic corresponds to a straightforward rule that fixes the power split at $u(t) \equiv 0.6$ and ignores battery status information. While the collocation solution provides a more "battery-friendly" strategy that is willing to burn more hydrogen, the baseline is a valuable reference point reflecting a "H_2-saving / battery-consuming" strategy, even though it is not the optimizer of the same cost functional.

Figure 2 and Tables 1 and 2 show that moving from the baseline to the collocation solution increases the total

hydrogen mass from $m_{\mathrm{H_2,tot}} = 0.751\,\mathrm{kg}$ to $1.062\,\mathrm{kg}$ (about a 40% increase), but improves the minimum SOC from 0.135 to 0.555 and reduces the peak battery current from $84.4\,\mathrm{A}$ to $66.1\,\mathrm{A}$. In qualitative Pareto terms, the collocation point lies further toward the "battery protection" corner of the tradeoff curve: it sacrifices hydrogen while bringing both state-of-charge and current closer to desirable limits.

The cost definition used here,

$$J = \int_0^T \Big(w_H\, \dot{m}_{\mathrm{H_2}}(t) + w_I\, i_{\mathrm{bat}}^2(t) \Big)\, \mathrm{d}t, \tag{31}$$

makes it straightforward to reason about how alternative weight choices would shift the solution, even without exhaustively recomputing the optimum. Increasing w_H relative to w_I would penalize hydrogen consumption more strongly, encouraging the optimizer to lean back toward battery usage: one would expect lower $m_{\mathrm{H_2,tot}}$ but deeper SOC swing and higher peak currents, moving the solution closer to the baseline on the SOC–current axes. Conversely, increasing w_I or tightening the current limit $i_{\mathrm{bat,max}}$ would push the optimizer toward even more conservative battery operation, with higher hydrogen usage but flatter SOC trajectories and reduced current ripple.

Regarding the terminal SOC target $\mathrm{SOC_{target}}$, a comparable sensitivity emerges. $\mathrm{SOC_{target}} = 0.6$ was selected as a representative "leave some charge for the next leg" constraint in the current study. The feasible set would be effectively expanded and the optimizer would be able to exchange some more depth of discharge for a slight decrease in hydrogen consumption if this criterion were loosened to, say, $\mathrm{SOC_{target}} = 0.5$. The solution would move in the opposite way if the target were tightened to $\mathrm{SOC_{target}} = 0.7$. This would force the fuel cell to carry more of the mission power, increasing $m_{\mathrm{H_2,tot}}$ while also protecting the battery.

This research concentrates on the single weight pair $(w_H, w_I) = (1, 10^{-4})$ and a fixed terminal SOC target due to runtime restrictions and the comparatively high cost of each collocation solve (Section 3.4). However, the cost functional's structure makes the qualitative impact of different weights and SOC targets clear, and the baseline and collocation solutions already distinguish two different operating philosophies. A more thorough trade-off characterization is left as an obvious extension for future study, such as calculating a Pareto front in the $(m_{\mathrm{H_2,tot}}, \max_t |i_{\mathrm{bat}}|)$ plane for many weight pairs.

3.4 Solver Performance and Comparison

Numerically speaking, the collocation-based optimal control issue and the baseline simulation test somewhat different parts of the tool chain. One forward integration of the stiff ODE system with `ode15s` is necessary for the baseline controller. The overall cost of one baseline simulation is low because, for the parameter set in Appendix B, `ode15s` completes the $T = 1800\,\mathrm{s}$ mission with no step rejections and time steps on the order of a few seconds.

On the other hand, when $N = 40$ segments are employed, the direct collocation formulation results in a restricted nonlinear program with $4(N + 1) = 164$ decision variables. Three initial-condition constraints and $3N = 120$ defect equations that enforce the trapezoidal discretization of the dynamics make up the equality constraints. The inequality constraints consist of the current limits $|i_{\mathrm{bat}}| \leq i_{\mathrm{bat,max}}$ enforced at each node and the terminal SOC condition $\mathrm{SOC}(T) \geq \mathrm{SOC_{target}}$. It is necessary to loop over all nodes, calculate mission power, battery current, and hydrogen consumption, and repeatedly invoke the right-hand side $f(x, u, t)$ in order to evaluate either the objective or the constraint function.

Table 2 summarizes the performance of `fmincon` on this collocation problem using the interior-point algorithm and default settings apart from the maximum number of iterations. The solver performs 19 major iterations and approximately 3.1×10^3 function evaluations before reaching the built-in limit on the maximum number of evaluations. At termination, the reported objective is $J \approx 1.1 \times 10^2$, the first-order

optimality measure is $\mathcal{O}(10^0)$, and the feasibility metric (maximum constraint violation) remains on the order of 7×10^2. In other words, the returned solution is not a fully converged optimum, but rather an approximate feasible solution obtained under a finite evaluation budget.

Table 2: Summary of `fmincon` performance on the collocation problem with $N = 40$ segments.

Quantity	Value
Number of decision variables	$4(N+1) = 164$
Number of dynamic defect equalities	$3N = 120$
Terminal and path inequalities	SOC terminal + $2(N+1)$ current bounds
Major iterations	19
Function evaluations (objective + constraints)	3135
Reported objective at termination	$J \approx 1.14 \times 10^2$
First-order optimality measure	≈ 4.1
Maximum constraint violation ("Feasibility")	$\approx 7.3 \times 10^2$
Termination reason	Max. function evaluations reached

As demonstrated in Figure 2 and measured in Table 1, the collocation solution significantly improves the state and control trajectories when compared to the baseline heuristic, despite the premature termination. This emphasizes a crucial practical point: even a partially converged collocation solution can produce a power split strategy that is qualitatively better than basic heuristics for moderate-size stiff optimal control problems, even though it requires many orders of magnitude more function evaluations than a single forward simulation. evaluations than a single forward simulation.

Because it uses `ode15s` as a black box and optimizes over a significantly smaller number of parameters (such as a few piecewise constant power-split levels), a shooting-based approach of the kind described in Section 2.6 would be less expensive per iteration from a methodological standpoint. However, these shooting techniques can have trouble enforcing state inequality requirements like SOC and current limits and are more susceptible to initial assumptions. Tighter control over constraint satisfaction and direct access to the state and control trajectories at the collocation nodes are traded for larger issue sizes and higher evaluation costs in this case.

The given solution should be regarded as an approximate optimum rather than a completely converged one since the interior-point approach ended at the function-evaluation limit with feasibility and optimality measures remaining of order 10^2. Although there are strong qualitative tendencies in battery stress and hydrogen utilization, minor numerical variations in the provided metrics shouldn't be taken too seriously.

3.5 Decision-Ready Summary

The baseline constant power-split strategy $u(t) \equiv 0.6$ for the representative mission profile and parameter set in Appendix B consumes a total hydrogen mass of $m_{\mathrm{H_2,base}} = 0.751\,\mathrm{kg}$, drives the battery down to a minimum state-of-charge of $\min_t \mathrm{SOC}(t) \approx 0.135$, and generates a peak current of $\max_t |I_{\mathrm{bat}}(t)| \approx 84.4\,\mathrm{A}$. Although this behavior is hydrogen-efficient, it is obviously aggressive in terms of battery health.

Using cost weights $(w_H, w_I) = (1, 10^{-4})$ and a terminal constraint $\mathrm{SOC}(T) \geq 0.6$, the direct collocation strategy produces an optimal trajectory that raises hydrogen consumption to $m_{\mathrm{H_2,opt}} = 1.062\,\mathrm{kg}$ (roughly 40% higher than the baseline), improves the minimum SOC to $\min_t \mathrm{SOC}(t) \approx 66.1\,\mathrm{A}$. To put it another way, the optimizer purposefully relies more on the fuel cell to maintain a healthier operating window for the battery.

From a decision-making standpoint, these findings indicate that operators who are primarily limited by battery life and thermal limits should favor strategies akin to the collocation solution and tolerate higher hydrogen consumption, whereas operators who are severely hydrogen-limited may favor a baseline-like strategy at the expense of deeper battery cycling. As a result, the trade-off between fuel consumption and battery stress is clear and measurable for the simplified model, directly addressing RQ1–RQ3 for this mission.

4 Discussion

4.1 Why Focus on Hydrogen Usage and Battery Current?

The fuel cell and battery are complementing components of a hydrogen-electric propulsion system. While the battery mostly manages *power transients* and limits thermal and lifetime performance, the fuel cell stack and its related tankage dominate the *mission energy* budget, driving fuel cost and aircraft range. Thus, it is sense to consider battery stress and hydrogen consumption as the two primary performance indicators for an energy management plan.

Total hydrogen consumption $m_{\mathrm{H_2,tot}}$ maps directly into required tank capacity and refuelling infrastructure. For regional aircraft concepts, differences of a few hundred grams of hydrogen per flight accumulate to substantial fuel and weight penalties over a fleet and over years of operation. In the baseline simulation the mission uses $m_{\mathrm{H_2,tot}} = 0.751\,\mathrm{kg}$, whereas the collocation strategy uses $1.062\,\mathrm{kg}$. This $\sim 40\%$ increase is significant at the system level, particularly if the trend persists across a range of missions.

Deep SOC excursions and high currents hasten battery deterioration and may exceed temperature and voltage restrictions. The essential physics that both huge currents and deep discharge are detrimental is captured by the simple current limit $|i_{\mathrm{bat}}| \leq i_{\mathrm{bat,max}}$ and SOC bounds used here, which are rough stand-ins for more complex aging models. With a peak current of $84.4\,\mathrm{A}$, the baseline controller drives the SOC down to $\mathrm{SOC}(T) \approx 0.135$, which is probably not suitable for repeated operation. In comparison, a qualitatively safer operating regime is represented by the collocation solution, which preserves $\mathrm{SOC}(T) \approx 0.555$ and lowers the peak current to $66.1\,\mathrm{A}$.

Concentrating the analysis on $m_{\mathrm{H_2,tot}}$,Therefore, $\min_t \mathrm{SOC}(t)$ and $\max_t |i_{\mathrm{bat}}(t)|$ offer a clear lens on the system tradeoffs: how much extra fuel must be burned to maintain the battery in a healthy operating window, and how optimal control techniques and stiff solvers can aid in exploring that trade space.

4.2 Modeling Choices and Their Implications

In the context of this course, the modeling decisions made in Section 2 balance tractable computing with physical fidelity. The battery is treated as a Thevenin equivalent with an affine open-circuit voltage–SOC relation and a single RC branch, while the fuel cell stack is represented by a static polarization curve with a straightforward first-order lag on power. When the model needs to be assessed thousands of times inside an optimizer, these decisions enable the system to be expressed as a three-state stiff ODE with a cost-effective right-hand side.

Many second-order effects are overlooked, which is a drawback. Fuel cell voltage is actually dependent on temperature, gas pressure, and water management; instead of a single time constant, transient dynamics may include compressor limits and air-path lags. Lithium-ion batteries also show temperature-dependent resistance, hysteresis, and rate-dependent capacity. Ignoring these effects tends to make the model more optimistic: the fuel cell can ramp and sustain power more readily, and the battery seems more resilient to

repeated high-current operation than it actually is.

The use of fixed converter efficiencies η_{fc} and η_{bat} is another significant simplification. In reality, these efficiencies change depending on the load and operating point, and the power consumption of extra parasitic loads (cooling fans, pumps, etc.) is dependent on both ambient conditions and P_{fc}. Because of these exclusions, the absolute values of peak currents and $m_{\mathrm{H_2,tot}}$ should not be considered design-ready figures. Rather, it is preferable to think of the current model as a *relative* comparison tool that shows trends: solutions that rely more on the battery use less hydrogen but run the danger of going against SOC and current limitations, and vice versa.

Despite these drawbacks, the lumped-parameter model worked well for testing stiff ODE solvers, collocation, and shooting concepts in a way that is both numerically challenging and sufficiently clear to allow for an analytical explanation of the primary tradeoffs.

4.3 Numerical Method Considerations

From a numerical methods viewpoint, this project illustrates how model stiffness and optimization structure influence solver choices. Even though the state dimension is small, the combination of an RC battery model and fuel cell power dynamics introduces time scales separated by roughly an order of magnitude. Explicit solvers such as `ode45` are capable of integrating the system, but they require prohibitively small step sizes during fast transients. The stiff solver `ode15s` is therefore a more natural choice: it automatically takes large steps in quasi-steady segments of the mission and tightens its step size near the power transitions.

The continuous-time optimum control issue is converted by the direct collocation method into a finite-dimensional nonlinear program that can be managed by `fmincon`. Tight connection between the optimizer and the ODE dynamics is advantageous because the optimizer can access state and control values at multiple points in time, and state inequality restrictions can be directly applied at the collocation nodes. Computational expense is a drawback; as seen in Section 3.4, a single collocation solve with $N = 40$ segments needed more than 3×10^3 function evaluations and still ended at the evaluation limit. To put it another way, collocation comes at a significant per-iteration cost in exchange for algorithmic convenience and constraint handling.

A shooting-based method, on the other hand, would optimize over a limited range of control parameters using the same stiff ODE solver, using only one call to `ode15s` for each evaluation. This lowers the cost per evaluation, but also complicates the enforcement of path constraints and makes the optimization landscape more nonlinear and sensitive to initial predictions. Ideas from both methods might be combined in a more production-focused setting: collocation in a smaller neighborhood to refine the solution and impose limits, and shooting for coarse parameter tweaking.

Compared to other MECH 568 topics, such as solving large linear systems with Gauss–Seidel, multigrid V-cycles, and LU decomposition in the diffusion homework, this project keeps the linear algebra "inside" MATLAB's solvers. However, the same themes appear: the tradeoff between direct (LU) and iterative (multigrid) methods is analogous to the tradeoff between explicit time-stepping and stiff solvers, and the tolerance and residual concepts used here are the same ones practiced in those assignments.

Overall, the project demonstrates that stiff ODE solvers and collocation can be applied successfully even in a relatively simple MATLAB implementation, but that careful attention must be paid to tolerance settings, grid resolution, and termination criteria to avoid overinterpreting partially converged solutions.

From a methods standpoint, the project can be viewed as applying the stiff, implicit time-stepping ideas from the Finite Difference and Time-Stepping lectures (Backward Euler, trapezoid/BDF families) to a lumped ODE model, rather than to a spatially discretized PDE.

4.4 Limitations and Future Work

A number of significant restrictions should be taken into consideration when interpreting the results shown here. Initially, the analysis concentrates on a single set of cost weights (w_H, w_I) and a single representative mission profile. Regional planes actually fly over a variety of routes, weather conditions, and air traffic restrictions. A more thorough analysis would calculate a Pareto front between hydrogen consumption and battery deterioration measures and investigate several mission profiles, possibly using Monte Carlo sampling.

Secondly, the component models are deterministic and simplified. Thermal dynamics are disregarded and fuel cell and battery characteristics are regarded as known constants. The model would become more realistic and the limitations on acceptable power split schemes would probably be tightened if temperature states, parameter uncertainty, and aging effects were included. Additionally, this would provide access to robust or stochastic optimum control formulations and uncertainty quantification, which are logical expansions of the computational techniques taught in this course.

Third, the current solution uses an offline, ”batch” approach to solve the optimal control problem. In actuality, receding-horizon control or a supervisory approach that responds to disruptions and model mismatch would be used to conduct aviation energy management. An intriguing next step would be to integrate the existing model into a model predictive control framework and evaluate its real-time viability, perhaps using a reduced-order or neural-network surrogate model.

Fourth, the numerical results show that function evaluation limits caused the collocation problem to be solved only roughly. To lessen the computational load and enhance convergence, future research could experiment with more sophisticated nonlinear programming solvers, adaptive collocation meshes, or analytic sensitivities. Even within MATLAB, the runtime may be greatly decreased by adjusting `fmincon` settings, investigating different techniques, or taking use of sparsity.

Lastly, even though this project focuses on stiff IVPs and optimal control, other MECH 568 tools, like the multigrid and Gauss–Seidel solvers created for elliptic problems in the homework, could be used to handle higher fidelity PDE-based models of the fuel cell and battery or to precondition or speed up the linear solves inside implicit time-stepping. Examining these extensions would improve the connection between the course subject and this project.

Despite these drawbacks, the project shows that standard numerical techniques and relatively compact stiff ODE models can already offer helpful insight into the tradeoffs inherent in hydrogen-electric propulsion, and it lays the groundwork for more in-depth modeling and control studies in subsequent work.

5 Conclusions

This project formulated and solved an optimal energy management problem for a simplified hydrogen–electric aircraft powertrain using tools from MECH 568. A stiff three-state ODE model of a PEM fuel cell and Thevenin battery was constructed and simulated with `ode15s` for a representative regional mission. A baseline constant power-split strategy was compared with a collocation-based optimal control strategy that trades hydrogen consumption against battery current while enforcing a terminal SOC constraint.

For the selected mission, the collocation strategy increases hydrogen consumption to $1.062\,\mathrm{kg}$ while maintaining $\mathrm{SOC}(T) \approx 0.555$ and lowering the peak current from approximately $84\,\mathrm{A}$ to $66\,\mathrm{A}$. In contrast, the baseline controller uses $0.751\,\mathrm{kg}$ of hydrogen but drives the battery to a minimum SOC of approximately 0.135. These findings show a definite, measurable trade-off between battery stress and fuel economy. The work shows numerically that stiff solvers and direct collocation may be successfully applied to such hybrid

systems in MATLAB, but it also emphasizes the significance of mesh resolution, evaluation limitations, and solver tolerances for interpreting partially converged optimization results. The project's ultimate goal of applying MECH 568 numerical approaches to address a relevant, real-world research topic is thus fulfilled.

Appendix A: MATLAB Code for Reproducibility

This appendix lists the single MATLAB script used to generate all simulation results and figures in this report. The file `MECH568_final_project.m` contains the main driver function and all helper routines as local functions in one place, so that the entire project can be reproduced by calling `MECH568_final_project` from the MATLAB command window. The script implements:

- The parameter setup and mission profile (`setup_params`, `mission_profile`);
- The hybrid fuel cell–battery dynamics (`hybrid_rhs`, `battery_current`, `fc_voltage`);
- Postprocessing to recover currents and hydrogen usage (`postprocess_currents`);
- The direct collocation optimal control formulation and interface to `fmincon` (`run_collocation`, `collocation_cost`, `collocation_constraints`, `unpack_z`).

For convenience the full script is reproduced below.

Listing 1: Main MATLAB script for baseline and collocation-based optimal power split.

```
function MECH568_final_project()
% MECH568_FINAL_PROJECT
% Baseline simulation for a hydrogen--electric aircraft powertrain.
%
% Single file with main driver and all helper functions as local functions.

  % ----------------- Setup and baseline configuration -----------------
  params = setup_params();

  % Baseline: fuel cell carries 60% of demanded power
  u0    = 0.6;
  u_fun = @(t) u0 * ones(size(t));

  % Initial conditions: SOC near upper bound, RC voltage = 0,
  % fuel cell power equal to baseline fraction of initial propulsive power.
  Pprop0 = mission_profile(0, params);
  x0 = [params.SOC0; 0; u0 * Pprop0];

  opts = odeset('RelTol', params.RelTol, ...
                'AbsTol', params.AbsTol, ...
                'MaxStep', 5.0);

  rhs = @(t,x) hybrid_rhs(t, x, u_fun(t), params);

  % ----------------- Integrate stiff ODE system -----------------
  [t, x] = ode15s(rhs, [0 params.T_final], x0, opts);

  SOC   = x(:,1);
  v_rc  = x(:,2); %#ok<NASGU>
  P_fc  = x(:,3);
  Pprop = mission_profile(t, params);
```

```
% ----------------- Post-process currents and hydrogen usage -----------------
[I_fc, mdot_H2, I_bat, P_bat] = postprocess_currents(t, x, u_fun, params); %#ok<ASGLU>

% Total hydrogen consumed [kg]
m_H2_tot = trapz(t, mdot_H2);

% Key metrics
I_bat_peak = max(abs(I_bat));
SOC_min    = min(SOC);

fprintf('Baseline results (u0 = %.2f):\n', u0);
fprintf('  Total H2 used: %.3f kg\n', m_H2_tot);
fprintf('  Peak |I_bat| : %.1f A\n', I_bat_peak);
fprintf('  Min SOC      : %.3f\n', SOC_min);

% ----------------- Collocation-based optimal control -----------------
[opt, t_opt, x_opt] = run_collocation(params, m_H2_tot, I_bat_peak, SOC_min);

% Recompute optimal currents and battery power for plotting
u_fun_opt = @(tt) interp1(opt.t_nodes, opt.u_nodes, tt, 'linear', 'extrap');
[~, ~, I_bat_opt, P_bat_opt] = postprocess_currents(t_opt, x_opt, u_fun_opt, params);
Pprop_opt = mission_profile(t_opt, params);
P_fc_opt  = x_opt(:,3);
SOC_opt   = x_opt(:,1);

% ----------------- Plots: baseline vs collocation-optimal -----------------
figure;

% Power
subplot(3,1,1);
plot(t,     Pprop/1e3, 'k-',  ...       % baseline Pprop
     t,     P_fc/1e3,  'b-',  ...       % baseline Pfc
     t,     P_bat/1e3, 'r-'); hold on;  % baseline Pbat
plot(t_opt, P_fc_opt/1e3,  'b--', ...   % optimal Pfc
     t_opt, P_bat_opt/1e3, 'r--');      % optimal Pbat
ylabel('Power [kW]');
legend('P_{prop} (baseline)', ...
       'P_{fc} (baseline)', 'P_{bat} (baseline)', ...
       'P_{fc} (optimal)',  'P_{bat} (optimal)', ...
       'Location','best');
grid on;

% SOC
subplot(3,1,2);
plot(t, SOC, 'k-', t_opt, SOC_opt, 'b--');
ylabel('SOC [-]');
legend('Baseline','Optimal','Location','best');
grid on;

% Battery current
subplot(3,1,3);
plot(t, I_bat, 'k-', t_opt, I_bat_opt, 'b--');
ylabel('I_{bat} [A]');
xlabel('Time [s]');
legend('Baseline','Optimal','Location','best');
grid on;
```

```

end  % ===== end main function =====


% ==================================================================
% Local functions
% ==================================================================

function params = setup_params()
%SETUP_PARAMS  Create parameter struct for hybrid H2-electric model.

  % Mission
  params.T_final   = 1800;                % [s]
  params.t_seg     = [0 300 1200 1500 1800];   % [s]
  params.P_seg_kw  = [80 50 30 40];       % [kW]

  % Fuel cell / hydrogen
  params.E0        = 240;                 % [V] open-circuit stack voltage
  params.A_act     = 0.06;                % [V] activation loss constant
  params.i_d       = 1;                   % [A] regularization current
  params.R_ohm     = 0.03;                % [ohm] ohmic resistance
  params.eta_fc    = 0.95;                % [-] electrical efficiency
  params.tau_fc    = 3;                   % [s] power time constant
  params.P_fc_min  = 5e3;                 % [W]
  params.P_fc_max  = 80e3;                % [W]

  % Hydrogen energetics (for mdot from power)
  params.LHV_H2    = 120e6;               % [J/kg]
  params.eta_stack = 0.6;                 % [-] overall stack + BoP efficiency

  % Battery / converter
  params.C_bat_As  = 40 * 3600;           % [A*s] 40 Ah
  params.V_OC_nom  = 400;                 % [V]
  params.k_OC      = 40;                  % [V] across SOC 0..1
  params.R0        = 0.05;                % [ohm]
  params.R1        = 0.02;                % [ohm]
  params.tau_rc    = 10;                  % [s]
  params.eta_bat   = 0.95;                % [-]
  params.SOC_min   = 0.2;
  params.SOC_max   = 0.9;
  params.SOC0      = 0.8;
  params.SOC_target = 0.6;
  params.I_bat_max = 200;                 % [A] current limit

  % Numerical tolerances
  params.RelTol    = 1e-6;
  params.AbsTol    = 1e-8;
end


function Pprop = mission_profile(t, params)
%MISSION_PROFILE  Prescribed propulsive power [W] as a function of time.

  t = t(:);
  PkW = zeros(size(t));

  for k = 1:length(params.P_seg_kw)
      t_start = params.t_seg(k);
      t_end   = params.t_seg(k+1);
```

```
        is_last = (k == length(params.P_seg_kw));

        % element-wise logical ops only
        in_seg = (t >= t_start) & ( (t < t_end) | (is_last & (t <= t_end)) );

        PkW(in_seg) = params.P_seg_kw(k);
    end

    Pprop = 1e3 * PkW;   % kW -> W
end


function dxdt = hybrid_rhs(t, x, u, params)
%HYBRID_RHS  Right-hand side for hybrid H2-electric model.
%
% State: x = [SOC; v_rc; P_fc].

    SOC   = x(1);
    v_rc  = x(2);
    P_fc  = x(3);

    % Mission and control
    Pprop = mission_profile(t, params);
    u     = max(0, min(1, u));   % clamp to [0,1]

    % Commanded FC power (apply bounds)
    P_fc_cmd = u * Pprop;
    P_fc_cmd = min(max(P_fc_cmd, params.P_fc_min), params.P_fc_max);

    % FC power dynamics
    dP_fc = (-P_fc + P_fc_cmd) / params.tau_fc;

    % Battery power from power balance (converter efficiency)
    P_bat = (Pprop - P_fc) / params.eta_bat;

    % Battery current from quadratic relation
    [I_bat, ~] = battery_current(P_bat, SOC, v_rc, params);

    % SOC dynamics
    dSOC = - I_bat / params.C_bat_As;

    % RC voltage dynamics
    dv_rc = (-v_rc + params.R1 * I_bat) / params.tau_rc;

    dxdt = [dSOC; dv_rc; dP_fc];
end


function I_fc = solve_fc_current(P_fc, params)
%SOLVE_FC_CURRENT  Solve for fuel cell current given power P_fc [W].
%
% (Still used to estimate current, but no longer used for H2 mass.)

    if P_fc <= 0
        I_fc = 0;
        return;
    end

```

```
  fun = @(i) params.eta_fc * fc_voltage(i, params) .* i - P_fc;

  % Initial guess: power / nominal voltage
  I_guess = max(1, P_fc / params.E0);

  I_fc = fzero(fun, I_guess);
end


function V = fc_voltage(i, params)
%FC_VOLTAGE  Fuel cell stack voltage from polarization relation.
  V = params.E0 ...
      - params.A_act .* log(i + params.i_d) ...
      - params.R_ohm .* i;
end


function [I_bat, V_bat] = battery_current(P_bat, SOC, v_rc, params)
%BATTERY_CURRENT  Solve quadratic for battery current given power.
%
% Uses Thevenin model:
%   V_bat = VOC(SOC) - R0*I_bat - v_rc
%   P_bat = V_bat * I_bat

  VOC = params.V_OC_nom + params.k_OC * (SOC - 0.5);

  % Quadratic: R0*I^2 - (VOC - v_rc)*I + P_bat = 0
  a = params.R0;
  b = -(VOC - v_rc);
  c = P_bat;

  disc = b.^2 - 4*a*c;
  disc = max(disc, 0);   % guard small negatives

  I1 = 0.5*(-b + sqrt(disc))/a;
  I2 = 0.5*(-b - sqrt(disc))/a;

  % Choose the physically reasonable root:
  % - For discharge (P_bat>0): both roots are positive; take the smaller one.
  % - For charge   (P_bat<0): roots have opposite sign; min(...) is negative.
  I_bat = min(I1, I2);

  % Enforce current limit
  I_bat = max(-params.I_bat_max, min(params.I_bat_max, I_bat));

  V_bat = VOC - params.R0 * I_bat - v_rc;
end


function [I_fc, mdot_H2, I_bat, P_bat] = postprocess_currents(t, x, u_fun, params)
%POSTPROCESS_CURRENTS  Recover currents and H2 usage from state history.

  SOC  = x(:,1);
  v_rc = x(:,2);
  P_fc = x(:,3);

  Pprop = mission_profile(t, params);
  u     = u_fun(t); %#ok<NASGU>

```

```
  % Fuel-cell current for each time point (for reporting only)
  I_fc  = zeros(size(t));
  for k = 1:length(t)
      if P_fc(k) <= 0
          I_fc(k) = 0;
      else
          I_fc(k) = solve_fc_current(P_fc(k), params);
      end
  end

  % Battery power and current
  P_bat = (Pprop - P_fc) / params.eta_bat;
  I_bat = zeros(size(t));
  for k = 1:length(t)
      [I_bat(k), ~] = battery_current(P_bat(k), SOC(k), v_rc(k), params);
  end

  % Hydrogen mass flow [kg/s] from power and LHV
  % mdot = P_fc / (eta_stack * LHV_H2)
  mdot_H2 = P_fc ./ (params.eta_stack * params.LHV_H2);
end

function [opt, t_opt, x_opt] = run_collocation(params, m_H2_base, I_bat_peak_base,
    SOC_min_base)
%RUN_COLLOCATION  Direct-collocation optimal power split using fmincon.
%
% Outputs:
%   opt   - struct with optimal metrics and power-split samples
%   t_opt - time vector for re-simulated optimal trajectory
%   x_opt - state history for optimal trajectory

  N = 40;                               % number of collocation segments
  nNodes = N + 1;
  T = params.T_final;
  t_nodes = linspace(0, T, nNodes)';

  % Mission power at nodes
  Pprop_nodes = mission_profile(t_nodes, params);

  % Initial guess: constant split u0 = 0.6 with simple state profiles
  u0 = 0.6;
  SOC_guess  = linspace(params.SOC0, max(params.SOC_target, params.SOC_min+0.05),
    nNodes)';
  v_guess    = zeros(nNodes,1);
  Pfc_guess  = u0 * Pprop_nodes;

  z0 = [SOC_guess; v_guess; Pfc_guess; u0*ones(nNodes,1)];

  % Variable bounds
  lb_SOC = params.SOC_min * ones(nNodes,1);
  ub_SOC = params.SOC_max * ones(nNodes,1);
  lb_v   = -1e3 * ones(nNodes,1);
  ub_v   =  1e3 * ones(nNodes,1);
  lb_Pfc = params.P_fc_min * ones(nNodes,1);
  ub_Pfc = params.P_fc_max * ones(nNodes,1);
  lb_u   = zeros(nNodes,1);
  ub_u   = ones(nNodes,1);

  lb = [lb_SOC; lb_v; lb_Pfc; lb_u];
```

```
  ub = [ub_SOC; ub_v; ub_Pfc; ub_u];

  % Cost weights
  w_H = 1.0;      % hydrogen mass
  w_I = 1e-4;     % battery current penalty

  cost    = @(z) collocation_cost(z, params, N, w_H, w_I);
  nonlcon = @(z) collocation_constraints(z, params, N);

  opts = optimoptions('fmincon', ...
      'Algorithm','interior-point', ...
      'Display','iter', ...
      'MaxIterations',80);

  fprintf('\n=== Starting collocation optimization ===\n');
  [z_opt, J_opt] = fmincon(cost, z0, [], [], [], [], lb, ub, nonlcon, opts);
  fprintf('Collocation optimum: J = %.3g\n', J_opt);

  % Unpack node values
  [SOC_opt_nodes, v_opt_nodes, Pfc_opt_nodes, u_opt_nodes] = unpack_z(z_opt, N); %#ok<ASGLU>

  % Re-simulate with optimal power split for clean metrics
  u_fun_opt = @(t) interp1(t_nodes, u_opt_nodes, t, 'linear', 'extrap');

  Pprop0 = mission_profile(0, params);
  x0_opt = [params.SOC0; 0; u_opt_nodes(1)*Pprop0];

  rhs_opt = @(t,x) hybrid_rhs(t, x, u_fun_opt(t), params);

  [t_opt, x_opt] = ode15s(rhs_opt, [0 params.T_final], x0_opt, ...
      odeset('RelTol',params.RelTol,'AbsTol',params.AbsTol,'MaxStep',5));

  [~, mdot_H2_opt, I_bat_opt, ~] = postprocess_currents(t_opt, x_opt, u_fun_opt, params);
  m_H2_opt       = trapz(t_opt, mdot_H2_opt);
  I_bat_peak_opt = max(abs(I_bat_opt));
  SOC_min_opt    = min(x_opt(:,1));

  fprintf('\nComparison: baseline vs collocation-optimal\n');
  fprintf('  H2 mass [kg]:   baseline = %.3f, optimal = %.3f\n', ...
          m_H2_base, m_H2_opt);
  fprintf('  peak |I_bat| A: baseline = %.1f, optimal = %.1f\n', ...
          I_bat_peak_base, I_bat_peak_opt);
  fprintf('  min SOC [-]:    baseline = %.3f, optimal = %.3f\n\n', ...
          SOC_min_base, SOC_min_opt);

  % Pack outputs for plotting / reporting
  opt.m_H2_opt        = m_H2_opt;
  opt.I_bat_peak_opt  = I_bat_peak_opt;
  opt.SOC_min_opt     = SOC_min_opt;
  opt.t_nodes         = t_nodes;
  opt.u_nodes         = u_opt_nodes;
end



function J = collocation_cost(z, params, N, w_H, w_I)
%COLLOCATION_COST  Objective: hydrogen mass + weighted battery current^2.
```

```

  nNodes = N + 1;
  T = params.T_final;
  dt = T / N;
  t_nodes = linspace(0, T, nNodes)';

  [SOC, v_rc, P_fc, ~] = unpack_z(z, N);

  Pprop_nodes = mission_profile(t_nodes, params);
  P_bat_nodes = (Pprop_nodes - P_fc) / params.eta_bat;

  I_bat_nodes   = zeros(nNodes,1);
  mdot_H2_nodes = zeros(nNodes,1);

  for k = 1:nNodes
      [I_bat_nodes(k), ~] = battery_current(P_bat_nodes(k), SOC(k), v_rc(k), params);
      mdot_H2_nodes(k)    = P_fc(k) / (params.eta_stack * params.LHV_H2);
  end

  % Trapezoidal integration of the running cost
  J = 0;
  for k = 1:N
      integrand_k   = w_H*mdot_H2_nodes(k)   + w_I*I_bat_nodes(k)^2;
      integrand_kp1 = w_H*mdot_H2_nodes(k+1) + w_I*I_bat_nodes(k+1)^2;
      J = J + 0.5*dt*(integrand_k + integrand_kp1);
  end
end


function [c, ceq] = collocation_constraints(z, params, N)
%COLLOCATION_CONSTRAINTS  Dynamics defects + SOC / current inequalities.

  nNodes = N + 1;
  T = params.T_final;
  dt = T / N;
  t_nodes = linspace(0, T, nNodes)';

  [SOC, v_rc, P_fc, u] = unpack_z(z, N);

  % --- Initial condition constraints ---
  Pprop0 = mission_profile(0, params);
  x0_desired = [params.SOC0; 0; u(1)*Pprop0];

  ceq_init = [SOC(1) - x0_desired(1);
              v_rc(1) - x0_desired(2);
              P_fc(1) - x0_desired(3)];

  % --- Dynamic defects (trapezoidal collocation) ---
  ceq_dyn = zeros(3*N,1);
  idx = 1;
  for k = 1:N
      tk    = t_nodes(k);
      tkp1  = t_nodes(k+1);

      xk    = [SOC(k);   v_rc(k);   P_fc(k)];
      xkp1  = [SOC(k+1); v_rc(k+1); P_fc(k+1)];

      fk    = hybrid_rhs(tk,   xk,   u(k),   params);
      fkp1  = hybrid_rhs(tkp1, xkp1, u(k+1), params);
```

```
        defect = xkp1 - xk - 0.5*dt*(fk + fkp1);
        ceq_dyn(idx:idx+2) = defect;
        idx = idx + 3;
    end

    ceq = [ceq_init; ceq_dyn];

    % --- Inequality constraints: terminal SOC and current limits ---
    % Terminal SOC: SOC_end >= SOC_target -> SOC_target - SOC_end <= 0
    c_soc_terminal = params.SOC_target - SOC(end);

    % Battery current limits at nodes
    Pprop_nodes = mission_profile(t_nodes, params);
    P_bat_nodes = (Pprop_nodes - P_fc) / params.eta_bat;
    I_bat_nodes = zeros(nNodes,1);
    for k = 1:nNodes
        [I_bat_nodes(k), ~] = battery_current(P_bat_nodes(k), SOC(k), v_rc(k), params);
    end

    c_Iplus  = I_bat_nodes - params.I_bat_max;
    c_Iminus = -I_bat_nodes - params.I_bat_max;

    c = [c_soc_terminal; c_Iplus; c_Iminus];
end


function [SOC, v_rc, P_fc, u] = unpack_z(z, N)
%UNPACK_Z  Split collocation decision vector into state and control blocks.

    nNodes = N + 1;
    SOC   = z(1:nNodes);
    v_rc  = z(nNodes+1:2*nNodes);
    P_fc  = z(2*nNodes+1:3*nNodes);
    u     = z(3*nNodes+1:4*nNodes);
end
```

Appendix B: Parameters and Mission Profile

B.1 Mission Power Profile

For all simulations I use a single representative regional mission of total duration $T = 1800\,\mathrm{s}$ (30 minutes). The propulsive power demand is prescribed as a piecewise-constant profile with four segments, roughly corresponding to climb, cruise, descent, and approach.

Table 3: Prescribed mission power profile $P_{\mathrm{prop}}(t)$.

Segment	Time interval (s)	Duration (s)	P_{prop} (kW)
1 (climb)	$0 \le t < 300$	300	80
2 (cruise)	$300 \le t < 1200$	900	50
3 (descent)	$1200 \le t < 1500$	300	30
4 (approach)	$1500 \le t \le 1800$	300	40

In the MATLAB implementation the mission is represented by the breakpoints $t_{\text{seg}} = [0, 300, 1200, 1500, 1800]$ and the corresponding power levels $P_{\text{seg}} = [80, 50, 30, 40]$ kW, and interpolated as a piecewise-constant function.

B.2 Fuel Cell and Hydrogen Parameters

The PEM fuel cell stack is modelled using the polarization relation in equation (5) with the parameters listed in Table 4. These values are representative rather than tied to a specific commercial stack.

Table 4: Fuel cell and hydrogen parameters.

Quantity	Symbol	Value
Open-circuit voltage (stack)	E_0	$240\,\text{V}$
Activation loss constant	A	$0.06\,\text{V}$
Regularization current	i_{d}	$1\,\text{A}$
Ohmic resistance (stack)	R_{ohm}	$0.03\,\Omega$
Fuel cell efficiency	η_{fc}	0.95
Fuel cell power time constant	τ_{fc}	$3\,\text{s}$
Min / max fuel cell power	$P_{\text{fc,min}}$, $P_{\text{fc,max}}$	$5\,\text{kW}$, $80\,\text{kW}$
Hydrogen molar mass	M_{H_2}	$2.0 \times 10^{-3}\,\text{kg/mol}$
Faraday constant	F	$96\,485\,\text{C/mol}$
H_2 lower heating value	LHV_{H_2}	$120\,\text{MJ/kg}$

B.3 Battery and Converter Parameters

The battery is represented by a Thevenin model with one RC branch as in equations (11)–(15). Table 5 summarizes the parameters.

Table 5: Battery and converter parameters.

Quantity	Symbol	Value
Nominal capacity	C_{bat}	$40\,\text{Ah}$
Effective capacity	C_{bat}	$40 \times 3600 = 144\,000\,\text{A s}$
Nominal open-circuit voltage	$V_{\text{OC,nom}}$	$400\,\text{V}$
OCV slope vs. SOC (linear model)	k_{OC}	$40\,\text{V}$ over SOC $[0, 1]$
Series resistance	R_0	$0.05\,\Omega$
RC branch resistance	R_1	$0.02\,\Omega$
RC time constant	τ_{rc}	$10\,\text{s}$
Converter / inverter efficiency	η_{bat}	0.95
SOC lower / upper bounds	SOC_{min}, SOC_{max}	0.2, 0.9
Initial SOC	SOC_0	0.8
Target terminal SOC	$\text{SOC}_{\text{target}}$	0.6
Max battery current magnitude	$i_{\text{bat,max}}$	$200\,\text{A}$

In the simulations the open-circuit voltage is approximated as a simple linear function of SOC,

$$V_{\text{OC}}(\text{SOC}) = V_{\text{OC,nom}} + k_{\text{OC}}\big(\text{SOC} - 0.5\big), \tag{32}$$

clipped as needed to remain within a physically reasonable range.

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